

Brainstorming & Ideation

Phase 1 of 8 - Project Documentation

Project: OptiCrop: Smart Agricultural Production Optimization Engine

Problem Space Exploration

Farmers often struggle to select the most suitable crop for their land, since the right choice depends on a complex combination of soil nutrients (Nitrogen, Phosphorous, Potassium), climate (temperature, humidity, rainfall), and soil pH. Poor crop selection leads to reduced yields, wasted resources, and financial loss.

The team explored how machine learning could turn easily-measurable soil and environmental readings into a concrete, actionable crop recommendation - reducing the guesswork currently involved in crop planning.

Ideas Considered

- Unsupervised clustering alone (grouping similar growing conditions) versus a supervised classifier that outputs a specific, named crop - a supervised classifier was chosen as the primary recommendation engine, with clustering used as a complementary exploratory analysis step.
- Comparing several classification algorithms (KNN, Decision Tree, Random Forest, Logistic Regression) - Logistic Regression was selected as the deployed model for its strong accuracy and simplicity, while clustering (K-Means) was retained to surface natural groupings among crops.
- A desktop tool versus a web application - a Flask-based web app was chosen for accessibility across devices without requiring local installation.
- Single-page form versus a multi-page site with Home / About / Find Your Crop sections - the multi-page structure was chosen to separate the project introduction from the interactive prediction tool.

Chosen Direction

OptiCrop combines K-Means Clustering (to explore natural crop groupings by environmental similarity) with a Logistic Regression classifier (to directly recommend the best crop for a given set of soil and climate readings). The system is deployed as a three-page Flask web application: Home, About, and Find Your Crop.